

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <math.h>
```

```
#define ROWS 30
```

```
#define COLS 60
```

```
char canvas[ROWS][COLS];
```

```
// Initialize canvas with '_'
```

```
void initCanvas() {
```

```
    for (int i = 0; i < ROWS; i++) {
```

```
        for (int j = 0; j < COLS; j++) {
```

```
            canvas[i][j] = '_';
```

```
        }
```

```
    }
```

```
}
```

```
// Display canvas
```

```
void displayCanvas() {
```

```
    for (int i = 0; i < ROWS; i++) {
```

```
        for (int j = 0; j < COLS; j++) {
```

```
            printf("%c ", canvas[i][j]);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

```
// Draw line (Bresenham-like simple version)
```

```

void drawLine(int x1, int y1, int x2, int y2) {
    int dx = abs(x2 - x1);
    int dy = abs(y2 - y1);

    int steps = (dx > dy) ? dx : dy;

    float xInc = (float)(x2 - x1) / steps;
    float yInc = (float)(y2 - y1) / steps;

    float x = x1, y = y1;

    for (int i = 0; i <= steps; i++) {
        if (x >= 0 && x < ROWS && y >= 0 && y < COLS)
            canvas[(int)x][(int)y] = '*';
        x += xInc;
        y += yInc;
    }
}

```

// Draw rectangle

```

void drawRectangle(int x, int y, int h, int w) {
    drawLine(x, y, x, y + w);
    drawLine(x, y, x + h, y);
    drawLine(x + h, y, x + h, y + w);
    drawLine(x, y + w, x + h, y + w);
}

```

// Draw triangle (simple right triangle)

```

void drawTriangle(int x, int y, int h) {

```

```

drawLine(x, y, x + h, y);    // vertical
drawLine(x, y, x + h, y + h); // diagonal
drawLine(x + h, y, x + h, y + h); // base
}

// Draw circle (simple midpoint approximation)
void drawCircle(int cx, int cy, int r) {
    for (int i = 0; i < ROWS; i++) {
        for (int j = 0; j < COLS; j++) {
            int dist = (i - cx) * (i - cx) + (j - cy) * (j - cy);
            if (dist <= r * r) {
                canvas[i][j] = '*';
            }
        }
    }
}

```

```

// Clear and redraw canvas
void clearCanvas() {
    initCanvas();
}

```

```

// Menu
int main() {
    int choice;

    initCanvas();

    while (1) {

```

```
printf("\n===== 2D GRAPHICS EDITOR =====\n");
printf("1. Draw Line\n");
printf("2. Draw Rectangle\n");
printf("3. Draw Triangle\n");
printf("4. Draw Circle\n");
printf("5. Display Canvas\n");
printf("6. Clear Canvas\n");
printf("7. Exit\n");
printf("Enter choice: ");
scanf("%d", &choice);
```

```
int x, y, x2, y2, h, w, r;
```

```
switch (choice) {
    case 1:
        printf("Enter x1 y1 x2 y2: ");
        scanf("%d %d %d %d", &x, &y, &x2, &y2);
        drawLine(x, y, x2, y2);
        break;

    case 2:
        printf("Enter x y height width: ");
        scanf("%d %d %d %d", &x, &y, &h, &w);
        drawRectangle(x, y, h, w);
        break;

    case 3:
        printf("Enter x y height: ");
        scanf("%d %d %d", &x, &y, &h);
```

```
drawTriangle(x, y, h);
```

```
break;
```

```
case 4:
```

```
printf("Enter center x y and radius: ");
```

```
scanf("%d %d %d", &x, &y, &r);
```

```
drawCircle(x, y, r);
```

```
break;
```

```
case 5:
```

```
displayCanvas();
```

```
break;
```

```
case 6:
```

```
clearCanvas();
```

```
printf("Canvas cleared!\n");
```

```
break;
```

```
case 7:
```

```
exit(0);
```

```
default:
```

```
printf("Invalid choice!\n");
```

```
}
```

```
}
```

```
return 0;
```

```
}
```